

****Summary of the Document****

The document discusses the concepts of formal languages and grammars, specifically focusing on the following topics:

Types of Grammars

- * **Type 0 (Unrestricted Grammar)**: A grammar with no restrictions on the production of strings.
- * **Type 1 (Context-Sensitive Grammar)**: A grammar where each production is of the form $A \rightarrow \alpha B \beta$, where A and B are non-terminals and α and β are strings of terminals and non-terminals.
- * **Type 2 (Context-Free Grammar)**: A grammar where each production is of the form $A \rightarrow \alpha$, where A is a non-terminal and α is a string of terminals and non-terminals.
- * **Type 3 (Regular Grammar)**: A grammar where each production is of the form $A \rightarrow aB$ or $A \rightarrow a$, where A and B are non-terminals and a is a terminal.

Equivalence of Grammars

- * **Linear Grammar (LG)**: A grammar where each production is of the form $A \rightarrow aB$ or $A \rightarrow a$, where A and B are non-terminals and a is a terminal.
- * **Right-Linear Grammar (RLG)**: A grammar where each production is of the form $A \rightarrow aB$, where A and B are non-terminals and a is a terminal.
- * **Left-Linear Grammar (LLG)**: A grammar where each production is of the form $A \rightarrow Ba$, where A and B are non-terminals and a is a terminal.

Simplification of Context-Free Grammars (CFGs)

- * **Removal of Useless Symbols**: A symbol is useless if it appears at least once in the derivation of a string and is not a non-terminal.
- * **Removal of Unit Productions**: A production is a unit production if it is of the form $A \rightarrow AB$, where A and B are non-terminals.
- * **Removal of ϵ -Productions**: A production is an ϵ -production if it is of the form $A \rightarrow \epsilon$, where A is a non-terminal.

Conversion to Chomsky Normal Form (CNF)

- * **Introduction of New Variables**: New variables are introduced to replace non-terminals that appear in the production of ϵ .
- * **Conversion of Productions**: Productions are converted to the form $A \rightarrow BC$, where A, B, and C are non-terminals.

Normal Forms

- * **Chomsky Normal Form (CNF)**: A grammar is in CNF if it has the following properties:
 - + All productions are of the form $A \rightarrow BC$ or $A \rightarrow a$, where A, B, and C are non-terminals and a is a terminal.
 - + There are no ϵ -productions.
 - + There are no unit productions.
- * **Greibach Normal Form (GNF)**: A grammar is in GNF if it has the following properties:
 - + All productions are of the form $A \rightarrow aB$ or $A \rightarrow a$, where A and B are non-terminals and a is a terminal.
 - + There are no ϵ -productions.
 - + There are no unit productions.